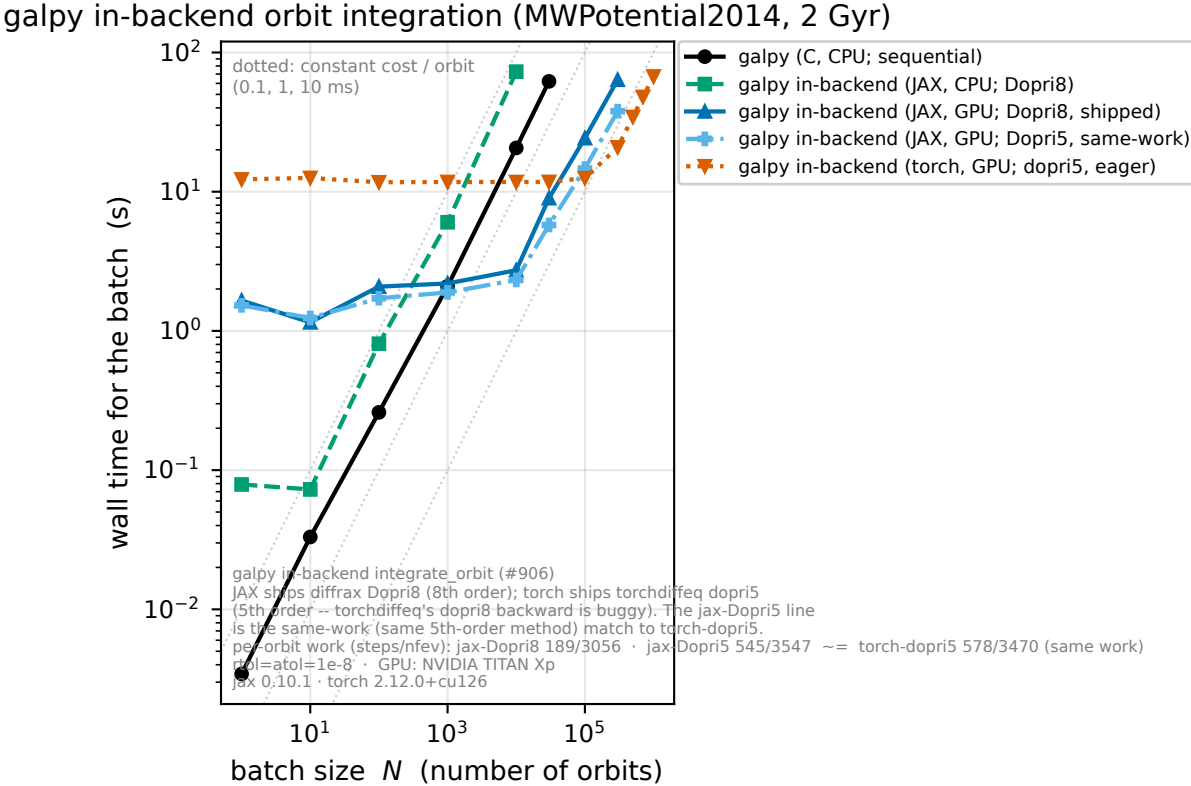
galpy in-backend orbit integration (MWPotential2014, 2 Gyr)
dotted: constant cost / orbit
(0.1, 1, 10 ms)
galpy (C, CPU; sequential)
galpy in-backend (JAX, CPU; Dopri8)
galpy in-backend (JAX, GPU; Dopri8, shipped)
galpy in-backend (JAX, GPU; Dopri5, same-work)
galpy in-backend (torch, GPU; dopri5, eager)
wall time for the batch (s)
batch size N (number of orbits)
galpy in-backend integrate_orbit (#906)
JAX ships diffrax Dopri8 (8th order); torch ships torchdiffeq dopri5
(5th order -- torchdiffeq's dopri8 backward is buggy). The jax-Dopri5 line
is the same-work (same 5th-order method) match to torch-dopri5.
per-orbit work (steps/nfev): jax-Dopri8 189/3056 · jax-Dopri5 545/3547 ~= torch-dopri5 578/3470 (same work)
rtol=atol=1e-8 · GPU: NVIDIA TITAN Xp
jax 0.10.1 · torch 2.12.0+cu126